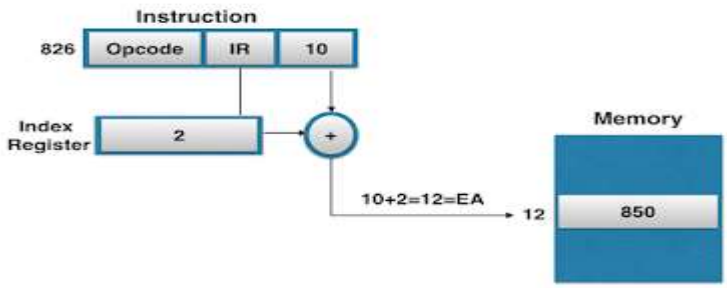
JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR
Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in
(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU & RTMNU

2 MARKS

1	Define number system.
2	Explain programmed IO.
3	What is direct mapping.
4	List out types of computer organization.
5	What is Cache memory.
6	Types of Interleaved Memory.
7	Concept of Pipelining.
8	Write DMA.
9	Give example of hardware and software.
10	Define Interleaved Memory.
11	Advantages of Direct-mapping.
12	Draw microinstruction code format.
13	Define RISC.
14	State the functional units of a basic computer system.
15	Explain Set-Associative Mapping.
16	Write Interleaved Memory.
17	Write characteristics of read only memory.
18	Give example of Direct Addressing Mode.
19	Define virtual memory.
20	Define Interrupt.
21	State the advantages of computer system.

22	Identify the following diagram and write correct name of addressing mode. 
23	List out types of addressing modes.
24	Write application of ROM..
25	List types of buses.
26	Define flash memory.
27	Give example of Register Addressing Mode.
28	Define decimal number system.
29	Draw the flow of Instruction cycle

4 MARKS

1	Compare virtual memory and cache memory.
2	Explain different addressing modes in Computer Architecture
3	Explain hardwired and microprogrammed control unit.
4	Examine X=(A+B)*(C-D)/E using 3, 2 and 0 Address Instruction format.
5	Write short note on Booth's Algorithm.
6	Draw single and double precision format of IEEE floating point representation.
7	Compare RAM and ROM
8	Explain Format of Microinstruction in Computer Architecture
9	Express the term Addressing modes, Interrupts, processor bus and interfacing.
10	Examine $(11011)_2 – (10011)_2$ using 1's complement and 2's complement method.
11	Draw Booth's Algorithm flowchart.
12	Difference between primary and secondary memory.
13	Explain static and dynamic RAM.
14	Discuss the following terms Memory delays, Branch delays, processor bus and interfacing.
15	Examine addition of (+15) and (-6) using 2's complement method.
16	Write short note on format of Microinstruction in Computer Architecture.
17	Draw typical CPU organization.

7 MARKS

1	Explain memory management requirements.
2	Explain non restoring algorithm with example.
3	What is RISC & CISC with characteristics.
4	multiplication of two signed number $(10111)_2$ & $(10011)_2$ by suitable algorithm

5	Discuss parallel processor system in computer architecture.
6	Explain the concept of pipelining. How it improves the execution of program over sequential execution?
7	What is programmed IO, IO modules.
8	Explain Fixed point representation with suitable example.
9	Examine (1269.14) in a Single precision and Double precision format.
10	What is Single Bus Architecture.
11	Analyse (703.63) with Single precision and Double precision format.
12	Examine $(1A5DB)_{10}$ to binary, octal and hexadecimal number system.
13	Discuss functioning of microprogrammed control unit..

12 MARKS

1	Rewrite about following: a) Programmed IO. b) Vector processing. c) IO modules. d) Control unit e) Types of parallel processor systems. f) Parallel Processing. g) Pipelining issues h) Microinstruction sequencing techniques. i) Displacement addressing mode. j) Superscalar processor
2	Evaluate division of 17 and 3 by using restoring algorithm.
3	Describe Direct-mapping, Associative Mapping & Set-Associative Mapping.
4	Multiply the following pair of signed 2's complement number using Booth's multiplication and bit pair recording technique A = 010111, B = 110110 A = Multiple & B = Multiplies.
5	Explain non-restoring algorithm. Find $(111)_2$ divided by $(110)_2$ by using Non-Restoring algorithm for unsigned integer.
6	Multiply -7 and +3 using Booth;s Algorithm with register size 4